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EX NO: 1	**Simulation of FDMA in MATLAB**

Aim: To simulate a Frequency Division Multiple Access (FDMA) system using MATLAB

and analyze its communication performance in terms of channel bandwidth, data rate, and signal-to-noise ratio (SNR).

Objective 1:

To simulate an FDMA system by allocating individual frequency channels to multiple users

and determine the channel bandwidth assigned to each user.

Output Parameter: Channel Bandwidth (kHz)

Procedure :

1. Open MATLAB and create a new script file.

2. Define the total available bandwidth, number of FDMA users, carrier frequencies, signal

power, and noise power.

3. Allocate a unique frequency channel to each user and calculate the channel bandwidth.

4. Compute the data rate for each user based on the allocated bandwidth and modulation

scheme.

5. Calculate the Signal-to-Noise Ratio (SNR) using the signal and noise power values.

6. Plot the transmitted signals of all FDMA users using different carrier frequencies.

7. Display the calculated channel bandwidth, data rate, and SNR in the MATLAB

Command Window.

8. Analyze the results to verify independent channel allocation and evaluate the overall

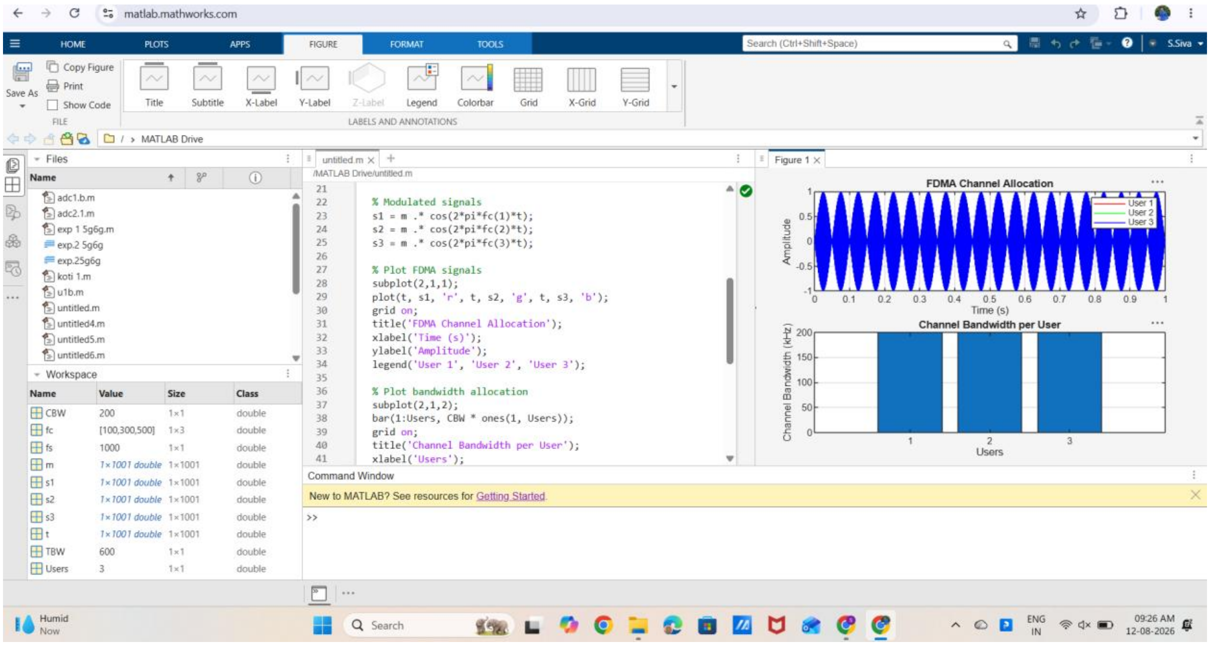
performance of the FDMA system.

Program (Matlab Code) :

```
clc; clear; close all;
fs=1000; t=0:1/fs:1;
TBW=600; Users=3;

CBW=TBW/Users;
fc=[100 300 500];
m=sin(2*pi*10*t);
s1=m.*cos(2*pi*fc(1)*t);
s2=m.*cos(2*pi*fc(2)*t);
s3=m.*cos(2*pi*fc(3)*t);
subplot(2,1,1)
plot(t,s1,'r',t,s2,'g',t,s3,'b'),grid on
title('FDMA Channel Allocation'),xlabel('Time (s)'),ylabel('Amplitude')
subplot(2,1,2)
bar(1:Users,CBW*ones(1,Users)),grid on
xlabel('Users'),ylabel('Channel Bandwidth (kHz)'),title('Channel Bandwidth per User')
```

OUTPUT:



Result:

Displays three users transmitting on different carrier frequencies (100 kHz, 300 kHz, and 500 kHz).

• Demonstrates that each user occupies a separate frequency channel.